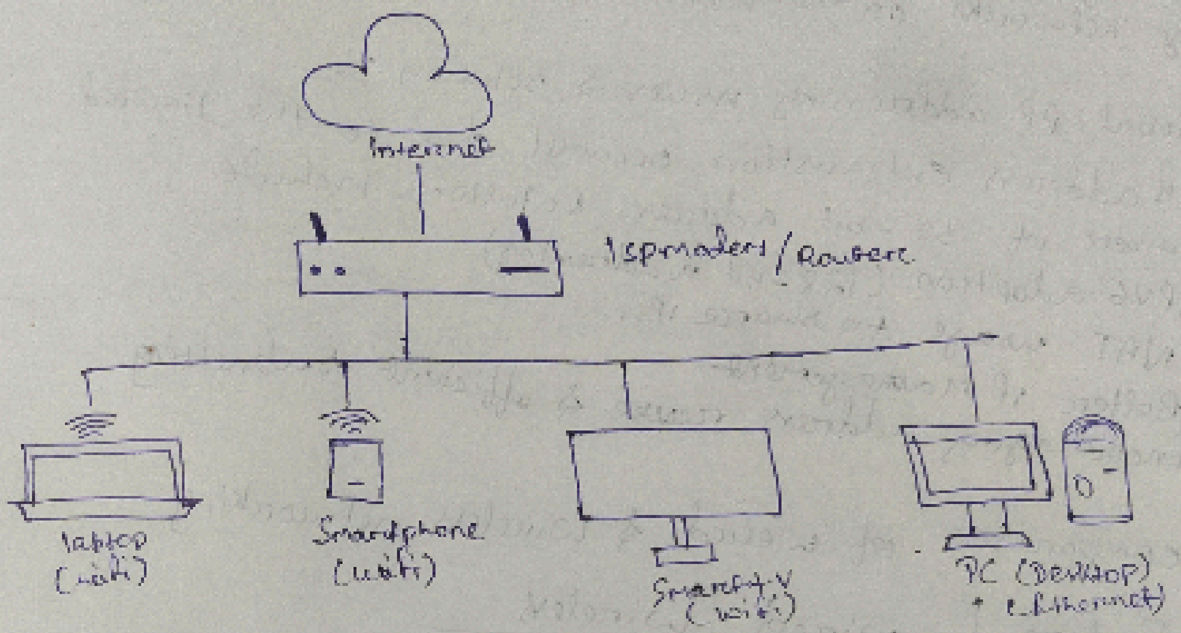
Assignment :- 1

Home Network Topology



i) IPV4 vs IPV6

- IPV4 uses 32-bit address, allowing ~4.3 billion address
- IPV6 uses 128-bit address, allowing trillions of devices
- IPV6 solves the address exhaustion problem and offers better Security and routing efficiency

ii) Subnetting basics:-

Subnetting divides a large network into smaller subnetworks (subnets). It helps manage IP association, improves security, and reduces congestion

iii) Network address translation

NAT allows multiple devices in a private network to access the internet using a single public IP address. It Maps private IPs to the public IP at the router level, conserving IPV4 addresses and adding a layer of security.

iv) Advanced IP addresses Management strategies:-

These include using DHCP with reservation, IPAM (Address management) tools, subnet planning & IP usage monitoring. They help in organizing large scale networks & avoiding it consists.

vi) Autonomous System (AS) numbers & their importance:-
An (AS) is a network or group of networks under one organization are crucial for BGP routing between large networks on the internet

vii) Global IP addressing issues & solution :-
IPV4 address exhaustion occured due to the limited Number of 32-bit address Solutions include
- IPV6 adoption (128-bit addressing)
- NAT usage to share IPs.
- Better IP Management
- Encouraging address reuse & efficient subnetting

viii) comparison of wired & wireless networking

Feature	wired	wireless
Speed	Generally faster	Slower than wired
Reliability	High	Susceptible to drops
Mobility	limited	Excellent
Setup	More complex	Easier

ix) wifi, Ethernet & emerging technologies
- wifi :- wireless networking using radion waves
- Ethernet : wired LAN connection, stable & fast
- Emerging ; wifi-7, 5G Integration.

x) Powerline & optical fiber communication
- Powerline : uses electrical wiring to transmit data ; useful enhance ~~courage & speed~~ where wifi is weak
- optical fibre ; uses light signals through glass cables ; offers ultra-high speed & long distant reliability.

Advanced Wireless technologies

xi) LIH uses LED light to transmit data at high speed it is secure, fast doesnt interfere with RF signals - ideal sensitive environment like hospitals.

xii) Performance Analysis & comparison of network types:-

- wired: offers consistent high speed & low latency - ideal for gaming streaming & work.
- wireless : offers flexibility & mobility but may suffer in speed & signal reliability.